VIGIL: Vision Gating with Intermittent Listening

Cost-Bounded Multimodal Perception over Continuous Video and Audio Streams

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Abstract

VIGIL — Vision Gating with Intermittent Listening. Multimodal large language models answer questions about video files, not about feeds that never end. A security camera, a wearable or a live broadcast produces an unbounded stream, and sending it to a model frame by frame makes context, cost and latency all grow without limit. VIGIL is a streaming pipeline that sits between a continuous audio-visual source and a commercial multimodal LLM API, and holds the cost of understanding that stream flat no matter how long it runs. Three mechanisms do the work: a zero-compute edge gate that discards visually static frames and silent audio before any encoder or network call; a bounded working memory that keeps a recent window verbatim and folds older history into a recursively rewritten summary; and a backpressure-controlled dispatch layer that matches ingest rate to generation latency. A cross-modal escalation rule, in which acoustic events raise the visual sampling rate, is evaluated as the project's specific contribution.

Keywords: streaming video understanding, multimodal large language models, edge filtering, bounded context, backpressure, cross-modal gating

1. Problem statement

Foundation multimodal models are built for files — a clip with a known start and end, uploaded once. A live feed has a start and no end. Three failures follow when a file-shaped pipeline is pointed at a feed:

1. **Unbounded context.** Attention cost rises quadratically with sequence length. An hour of video at 2 fps reaches roughly a million tokens, past the horizon most models were trained on.
2. **Redundant ingestion.** A corridor camera at night yields thousands of near-identical frames and hours of silence, all of it encoded and transmitted.
3. **Rate mismatch.** Video arrives at 25–30 fps; a multimodal API replies in seconds. Coupled directly, the queue grows at the difference until the buffer overflows or the provider returns HTTP 429.

The consequence is that no one runs a multimodal LLM on a camera for a week. This project asks what must sit between camera and model so that they can, and holds itself to one claim: **the sixth hour should cost what the first hour cost.**

2. Objectives

Four objectives reduce to three. The two that both addressed complexity — bounding context growth and decimating redundant input — are one objective, since they are two layers of a single goal: holding cost flat. Either alone still grows, because perfect filtering with append-only history still reaches quadratic cost, and perfect summarisation of an unfiltered feed still saturates bandwidth. The two that addressed responsiveness — decoupling perception from generation, and flow control — are likewise one, since backpressure is what makes decoupling work rather than a separate mechanism.

O1 — Hold the cost of understanding flat as the stream runs

Discard redundant visual and acoustic input before encoding, and keep working memory at fixed size, so resource use per hour does not grow with elapsed time.

O2 — Keep the pipeline responsive and stable under continuous load

Run perception and generation as independent asynchronous stages joined by a bounded queue, so a slow model response throttles capture instead of stalling the pipeline or exhausting quota.

O3 — Use the cheap modality to set the sampling rate of the expensive one

Let acoustic events raise the visual sampling rate for a bounded interval, so the gate reacts to onsets a pixel comparison cannot see.

	Measurable parameter	Method of measurement	Target
O1	Frame retention rate	Frames forwarded ÷ frames captured	≤ 20%
O1	Audio retention rate	Speech seconds ÷ seconds captured	≤ 30%
O1	Tokens per clock hour	Provider-reported usage	Hour 6 within ±15% of hour 1
O1	Context size	Tokens in dispatched prompt	Constant, ≤ fixed MaxLen
O1	Event recall	Events described ÷ events annotated	≥ 90% of ungated baseline
O2	End-to-end latency	Event onset → text emitted, p95	< 3 s
O2	Queue depth	Max pending items, 6-hour run	Bounded, no monotonic growth
O2	Rate-limit faults	HTTP 429 + quota errors per run	0
O2	Capture continuity	Uptime across source reconnects	≥ 99%
O3	Recall gain, audio-cued events	Escalated recall − visual-only recall	≥ +15 percentage points
O3	Escalation cost	Extra tokens ÷ baseline tokens	≤ +10%
O3	False escalation rate	Escalations with no event ÷ total	≤ 30%

3. Literature survey

The field moved decisively between 2024 and 2026: streaming multimodal understanding is now an active area with its own benchmarks. The survey is grouped by the three claims this project rests on, plus serving and evaluation.

A. Streaming versus offline architecture

Dispider (CVPR 2025) shows offline Video-LLMs requiring a whole file before inference fail in real time, and disentangles perception from decision and reaction. **Flash-VStream** (ICCV 2025) separates continuous frame encoding from query handling using a dual-tier memory combining a context synopsis with detail augmentation for high-entropy keyframes. **StreamMind** (ICCV 2025) introduces event-gated LLM invocation: a Cognition Gate between encoder and LLM invokes the model only when query-relevant events occur, raising throughput from a few fps to 100 fps. **FlowNar** (2026) has the LLM itself decide at each frame whether to narrate or emit a skip token.

B. Token decimation and input filtering

STC (CVPR 2026) caches features from temporally similar frames and prunes low-saliency tokens before prefill, cutting latency up to 45.3% while retaining about 99% accuracy. **FastV** (ECCV 2024) shows up to half of visual tokens can be pruned after layer 2; FastVID extends this temporally. **StreamMem** (2025) filters at the input: consecutive frames whose embedding cosine similarity exceeds a threshold are merged before the MLLM sees them. **DAT** (2026) is the closest systems analogue — an edge-cloud scheme where a lightweight small model screens at the edge to reduce large-model invocations, noting video already accounts for 76% of mobile data traffic.

C. Bounded memory over unbounded horizons

MemGPT (2024) applies OS virtual-memory paging to LLM context, with recursive summarisation under memory pressure. **StreamingLLM** (ICLR 2024) shows that retaining four attention-sink tokens alongside a sliding window stabilises perplexity over four million tokens at constant memory. **MovieChat** (CVPR 2024) consolidates dense visual tokens into sparse memory. **StreamBridge** (2025) applies round-decayed token merging to hold a fixed input length; **ReMem** (2026) adds retrieval because progressive merging loses early-frame detail; **CoRDS** (2026) replaces token-wise pruning with coreset selection under a fixed cache budget. A 2026 rate–distortion analysis quantifies the cost: irreversible summarisation loses roughly half its retained facts at every compaction event, a divergence invisible under single-turn evaluation.

D. Serving, flow control and load shedding

vLLM / PagedAttention (SOSP 2023) and **Sarathi-Serve** (OSDI 2024) establish chunked, bounded scheduling; Sarathi interleaves chunked prefills with decoding to remove head-of-line blocking. **Utility-aware load shedding for edge video analytics** (2023) implements a token-based backpressure loop between a shedder and a backend executor, dropping lower-utility frames under pressure.

E. Benchmarks and evaluation

StreamingBench (2025) provides 900 videos and 4,500 QA pairs across 18 streaming tasks including proactive output. **StreamArena** (2026) targets hour-scale interactive streaming with long-horizon memory, observing that high scores on short-clip benchmarks do not establish sustained causal understanding. **Ego-Exo4D** (CVPR 2024) supplies 1,422 hours of synchronised egocentric and exocentric video with spatial audio. **XD-Violence** and **UCF-Crime** provide untrimmed audio-visual anomaly footage with event-level annotation.

4. Research gap

The gaps stated in the earlier draft of this project are no longer open, and claiming them would not survive review.

Earlier claim	Status as of October 2026
MLLMs are built for offline files, not live streams	**Closed.** Dispider, Flash-VStream, StreamMind and StreamBridge are all natively streaming
No bounded-state working memory exists	**Closed.** StreamingLLM, MemGPT, StreamBridge, ReMem and CoRDS all bound memory explicitly
Pipelines send raw frames without edge partitioning	**Closed.** STC caches similar frames; StreamMem merges on embedding similarity; DAT screens at the edge
No backpressure or rate governance in serving	**Closed.** Sarathi-Serve chunks prefills; utility-aware shedding exists for edge video analytics

What the survey does reveal is a shared assumption. **Every one of those methods requires white-box model access.** StreamMind's Cognition Gate reuses the LLM's own shallow layers. StreamMem, CoRDS and ReMem manipulate the KV cache directly. STC modifies ViT encoding and prefill. None of these are available behind a commercial HTTP endpoint, where the binding constraint is not GPU memory but tokens and requests per hour. An open vLLM RFC on bounded-memory video sessions confirms that the field still frames this as a KV-cache problem.

A second assumption follows from the first. **Every published gate still runs a neural encoder on every frame.** StreamMem computes embeddings for all frames in order to decide which to merge; StreamMind runs CLIP continuously. That implies a GPU per stream, indefinitely. A gate with no neural computation at all, cheap enough for a Raspberry Pi or a phone, is the only option at the camera itself, and its recall cost has not been characterised — because white-box work has no reason to measure it.

Gap addressed by this project: cost-bounded streaming perception for *black-box* multimodal APIs using *zero-compute* edge gating — and, within it, the specific open question of **cross-modal escalation**. Audio-visual work to date fuses modalities *inside* the model (AVadCLIP, router-gated AVSR, entropy-gated captioning) or prunes audio tokens after encoding (FastAV). Using a near-free acoustic trigger to govern the *visual sampling rate upstream of the encoder* is, as far as this survey found, unaddressed. That is the contribution O3 tests.

Boundary of the claim. This is a measurement and systems-integration contribution, not a new model or a new attention mechanism. Its appropriate venue is a workshop or IEEE conference paper, not a main-track vision conference.

5. Research methodology

Architecture — four stages

1. **Capture.** A reconnect-safe source reads an HLS or RTSP feed (resolved via yt-dlp for public 24/7 streams) at 360p, with a parallel FFmpeg process emitting raw PCM. Both stamp a shared wall clock so chunks stay aligned. Live HLS URLs expire within hours, so re-resolution and resumption are part of the design, not error handling.
2. **Gate (Level 0).** Per frame: grayscale downscale to 128×128, then SSIM against the last retained frame; forward only if SSIM < τ. Audio passes WebRTC VAD; non-speech segments are dropped. No neural computation and no GPU at this stage.
3. **Dispatch.** Retained frames and speech segments form bounded chunks on an `asyncio.Queue(maxsize=N)`. A worker pool with fixed `max_concurrency` and a sliding-window rate limiter consumes it. A full queue blocks capture — that is the backpressure.
4. **Memory.** The last k chunk descriptions are held verbatim; everything older is folded into a summary regenerated recursively at each compaction. An anchor block (task, scene, persistent entities) is pinned at the head of every request, following StreamingLLM's attention-sink finding.

Cross-modal escalation (O3)

On a VAD-positive segment whose RMS energy exceeds a threshold, τ is lowered for the next 30 seconds, raising the effective frame rate. The rule itself is a few lines of code; the experiment comparing it against visual-only gating is the contribution.

Datasets and sources

XD-Violence and UCF-Crime supply annotated audio-visual events with ground-truth timings. A self-recorded 6-hour corridor or desk feed covers the long-horizon runs, where no public set has ground truth at that duration. One public 24/7 traffic stream provides live-operation evidence.

Experimental design — four runs over identical footage

Run	Gate	Memory	Purpose
A — baseline	none	append-only	Reference recall; establishes the cost curve to beat
B — gated	SSIM + VAD	append-only	Isolates the gate's effect on recall and bandwidth
C — full	SSIM + VAD	bounded	Tests O1 and O2 together
D — escalated	SSIM + VAD + audio escalation	bounded	Tests O3 against run C

Protocol

Each run is 6 hours on the same recorded footage, replayed at real-time rate so timing behaviour is reproducible. Instrumentation logs per-chunk token usage, queue depth, dispatch latency and gate decisions to CSV. Recall is scored by matching emitted descriptions to annotated event windows, with a fixed matching window and a second annotator on a 10% sample to report agreement. A τ sweep across five values yields the recall-versus-tokens curve, which is the project's headline figure.

Threats to validity

Commercial API latency varies with provider load, so latency figures are reported as distributions across runs rather than single values. Self-recorded footage risks staging bias, so event timings are annotated by someone who did not record the footage.

6. Scope and deliverables

The scope is deliberately narrowed from the original specification. Earlier versions of this project (a full sandbox for multimodal agents, then a QEMU-based telemetry harness) required hypervisor work, causal DAG construction and copy-on-write checkpointing — none of which is buildable alongside coursework, and none of which is needed to test the objectives above.

In scope	Out of scope
Two input streams: one video, one audio	GUI / OS event capture, accessibility trees
SSIM + VAD gating, no model training	Fine-tuning or training any model
One swappable API backend, plus a local model for offline testing	Multi-provider routing, custom serving stack
Bounded queue, rate limiter, rolling summary	Checkpointing, state forking, time-travel debugging
Four comparative runs on recorded footage	Deployment, multi-camera scale-out
Dashboard showing live timeline and three counters	Production UI

Deliverables: a runnable pipeline; a live demo on a public stream or phone camera; the recall-versus-tokens curve across the τ sweep; the four-run comparison table; and this document.

Why generative AI is central. The chunk captions, the recursive summarisation that compresses history, and the event descriptions are all LLM calls. SSIM and VAD are classical signal processing acting as gatekeepers. That contrast is the project's argument: it is what makes continuous generative perception affordable.

7. References

1. R. Qian et al., "Dispider: Enabling Video LLMs with Active Real-Time Interaction via Disentangled Perception, Decision, and Reaction," CVPR 2025.
2. H. Zhang et al., "Flash-VStream: Efficient Real-Time Understanding for Long Video Streams," ICCV 2025.
3. X. Ding et al., "StreamMind: Unlocking Full Frame Rate Streaming Video Dialogue through Event-Gated Cognition," ICCV 2025.
4. Y. Wang et al., "Accelerating Streaming Video Large Language Models via Hierarchical Token Compression (STC)," CVPR 2026.
5. L. Chen et al., "FastV: An Image is Worth 1/2 Tokens After Layer 2," ECCV 2024.
6. "StreamMem: Query-Agnostic KV Cache Memory for Streaming Video Understanding," arXiv:2508.15717, 2025.
7. "DAT: Dual-Aware Adaptive Transmission for Efficient Multimodal LLM Inference in Edge-Cloud Systems," arXiv:2604.05375, 2026.
8. C. Packer et al., "MemGPT: Towards LLMs as Operating Systems," 2024.
9. G. Xiao, Y. Tian, B. Chen, S. Han and M. Lewis, "Efficient Streaming Language Models with Attention Sinks," ICLR 2024.
10. E. Song et al., "MovieChat: From Dense Token to Sparse Memory for Long Video Understanding," CVPR 2024.
11. "StreamBridge: Turning Your Offline Video Large Language Model into a Proactive Streaming Assistant," arXiv:2505.05467, 2025.
12. "ReMem: Streaming Video Understanding With Long Context Retention," arXiv:2610.05940, 2026.
13. "CoRDS: Coreset-based Representative and Diverse Selection for Streaming Video Understanding," arXiv:2605.14310, 2026.
14. W. Kwon et al., "Efficient Memory Management for Large Language Model Serving with PagedAttention," SOSP 2023.
15. A. Agrawal et al., "Sarathi-Serve: Efficient LLM Inference by Chunking Prefills," OSDI 2024.
16. "Utility-Aware Load Shedding for Real-time Video Analytics at the Edge," arXiv:2307.02409, 2023.
17. "StreamingBench: Assessing the Gap for MLLMs to Achieve Streaming Video Understanding," arXiv:2411.03628, 2025.
18. "StreamArena: Toward Continuous, Interactive, and Long-Horizon Agentic Streaming Video Understanding," arXiv:2608.05703, 2026.
19. K. Grauman et al., "Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives," CVPR 2024.
20. P. Wu et al., "Not only Look, but also Listen: Learning Multimodal Violence Detection under Weak Supervision (XD-Violence)," ECCV 2020.
21. "AVadCLIP: Audio-Visual Collaboration for Robust Video Anomaly Detection," arXiv:2504.04495, 2025.
22. "What to Keep, What to Forget: A Rate–Distortion View of Memory Compaction in LLMs and Agents," arXiv:2607.08032, 2026.